

FEATURE IMPORTANCE ANALYSIS REPORT

Feature Engineering Using Titanic Dataset

Machine Learning Model: Random Forest Classifier

Feature Engineering Analysis

1. Project Overview

The objective of this project is to demonstrate how feature engineering can create meaningful new variables from a structured dataset and potentially improve machine learning model performance. The Titanic passenger dataset was selected as the structured public dataset. The analysis includes family-based features, categorical binning, interaction features, and date decomposition. A Random Forest classifier was trained before and after feature engineering to compare predictive performance. Feature importance was then analyzed to identify the variables that contributed most strongly to the engineered model.

Engineered Features

1. FamilySize
2. IsAlone
3. FarePerPerson
4. AgeGroup
5. FamilyCategory
6. Sex_Pclass (interaction feature)
7. VoyageYear
8. VoyageMonth
9. VoyageDay
10. VoyageDayOfWeek

Note on Date Decomposition

The Titanic dataset does not contain a passenger-specific date column. Therefore, the historical Titanic voyage date was used as a reference date to demonstrate date decomposition into year, month, day, and day-of-week components.

2. Model Performance Comparison

Performance Before and After Feature Engineering

Metric	Before	After
Accuracy	0.8101	0.8045
Precision	0.7778	0.7500
Recall	0.7101	0.7391
F1 Score	0.7424	0.7445

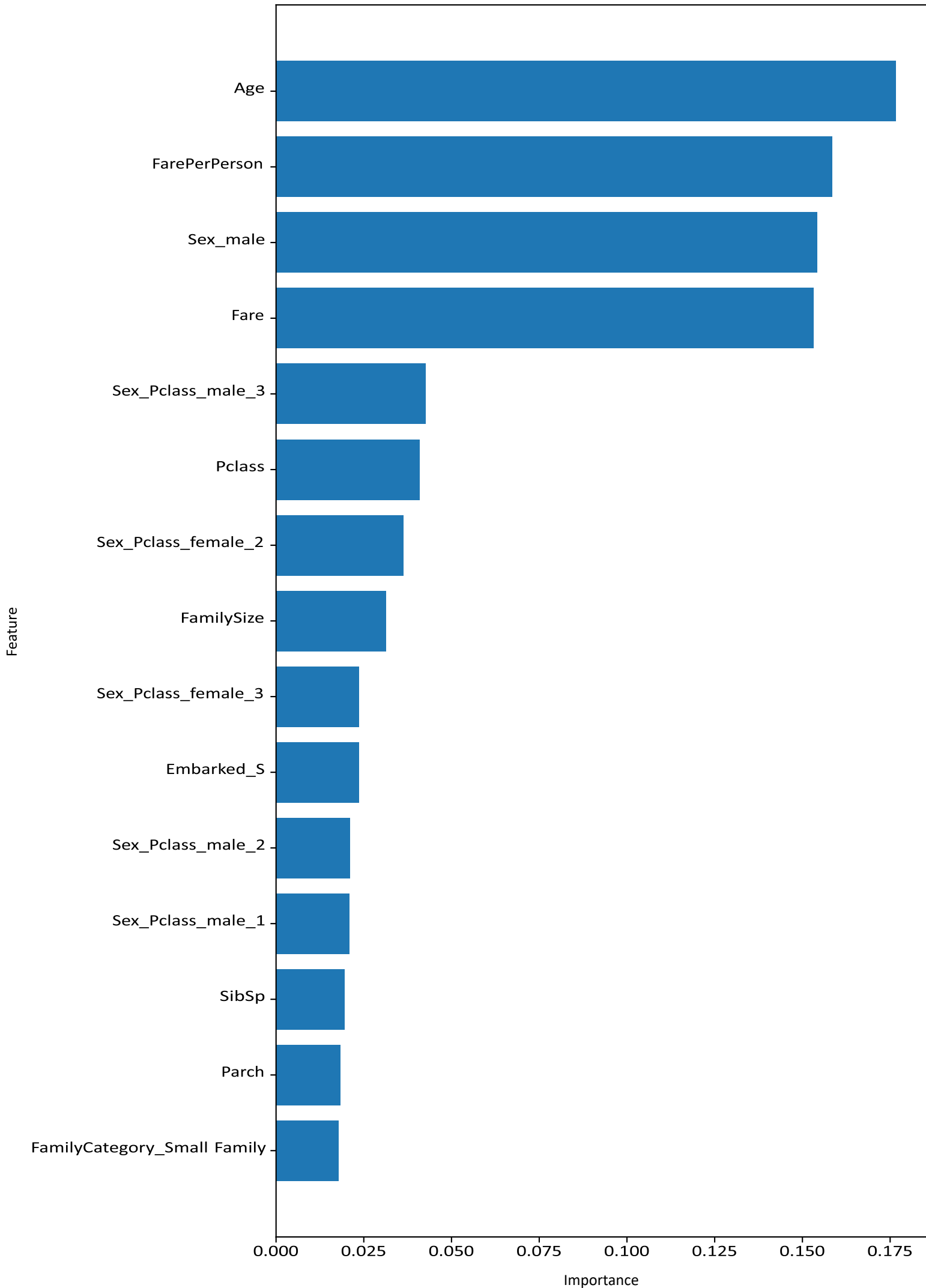
Accuracy Improvement

-0.56 percentage points

F1 Score Improvement

0.21 percentage points

Top 15 Feature Importances



3. Feature Importance Ranking

Rank	Feature	Importance (%)
1	Age	17.66
2	FarePerPerson	15.85
3	Sex_male	15.43
4	Fare	15.31
5	Sex_Pclass_male_3	4.26
6	Pclass	4.09
7	Sex_Pclass_female_2	3.63
8	FamilySize	3.13
9	Sex_Pclass_female_3	2.37
10	Embarked_S	2.36

Interpretation

Feature importance indicates how strongly each input variable contributes to the predictions made by the Random Forest model. Features with larger importance values have a greater influence on the model's decision-making process. Engineered variables can therefore be compared with the original variables to determine whether the feature engineering process introduced useful predictive information.

4. Conclusion

Feature engineering was applied to the Titanic dataset to create additional variables representing family structure, fare distribution, age groups, categorical interactions, and date components. The Random Forest classifier was evaluated before and after feature engineering. The baseline model achieved an accuracy of 0.8101, while the engineered model achieved an accuracy of 0.8045. The baseline F1 score was 0.7424, compared with 0.7445 after feature engineering. The feature-importance analysis provides insight into which variables contributed most strongly to the final model. Overall, the experiment demonstrates how transforming raw variables into meaningful features can provide additional information for predictive modelling.